

MMRM: A Multiplex Multimodal Representation Model for Product Ranking in E-commerce Search

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Abstract

Multimodal information is pivotal for e-commerce search ranking. Existing works leverage multimodal data typically by fine-tuning general Multimodal Large Language Models (MLLMs) via collaborative signals, subsequently integrating the derived representations into ranking models as item features. Despite their efficacy, these methods face two primary limitations: (1) they rely on a single collaborative signal for MLLM fine-tuning, failing to exploit the heterogeneous signals essential for multitask ranking; and (2) they treat multimodal representations as regular item features in ranking models, underutilizing their latent potential for user behavior modeling. To address these challenges, we propose the Multiplex Multimodal Representation Model (MMRM), a unified framework that aligns MLLMs with diverse collaborative signals. By employing a shared backbone with task-specific tokens and projection layers, MMRM simultaneously learns from multiple signals and generates comprehensive multiplex item representations in a single inference pass. Furthermore, we introduce a multiplex user representation strategy in ranking models, which derives task-specific user representations via search-based behavior sequence modeling leveraging multiplex item representations. Extensive experiments demonstrate MMRM's superior efficiency and effectiveness. Notably, MMRM has been successfully deployed in the JD e-commerce search engine, yielding significant performance gains for millions of daily users.

CCS Concepts

• Information systems → Information retrieval.

Keywords

Multimodal Representation, Contrastive Learning, Multitask Learning, E-commerce Search System

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1 Introduction

E-commerce search engines rely on ranking models to capture fine-grained user preferences across multiple objectives, such as Click-Through Rate (CTR), Add-to-Cart Rate (ACR), and Conversion Rate (CVR). Conventional ID-based ranking models inherently suffer from data sparsity and an inability to perceive product semantics (e.g., titles and images) [17, 25]. To address this, recent works fine-tune general Multimodal Large Language Models (MLLMs) using collaborative signals from e-commerce, then integrate the resulting representations into ranking models as auxiliary features [10, 11].

Despite these advancements, two limitations persist. First, existing methods rely on a single collaborative signal for MLLM fine-tuning, such as search-based query-to-item (q2i) or behavior-based item-to-item (i2i) clicks [13, 20, 26]. We argue that e-commerce signals are inherently heterogeneous: while q2i clicks capture broad relevance (Fig. 1a-c), sequential i2i behaviors—especially cart and order signals—reflect stronger fine-grained similarities and complementary relationships (Fig. 1d-g). Neglecting these diverse signals prevents MLLMs from learning comprehensive representations essential for multitask ranking. Second, multimodal representations are underutilized in downstream ranking models. Most ranking models introduce one multimodal embedding table for hand-crafted feature interactions [20, 24] or simple behavior grouping [21, 23], failing to capture multiplex user intents. This leads to information loss and embedding entanglement in multitask scenarios [6, 18].



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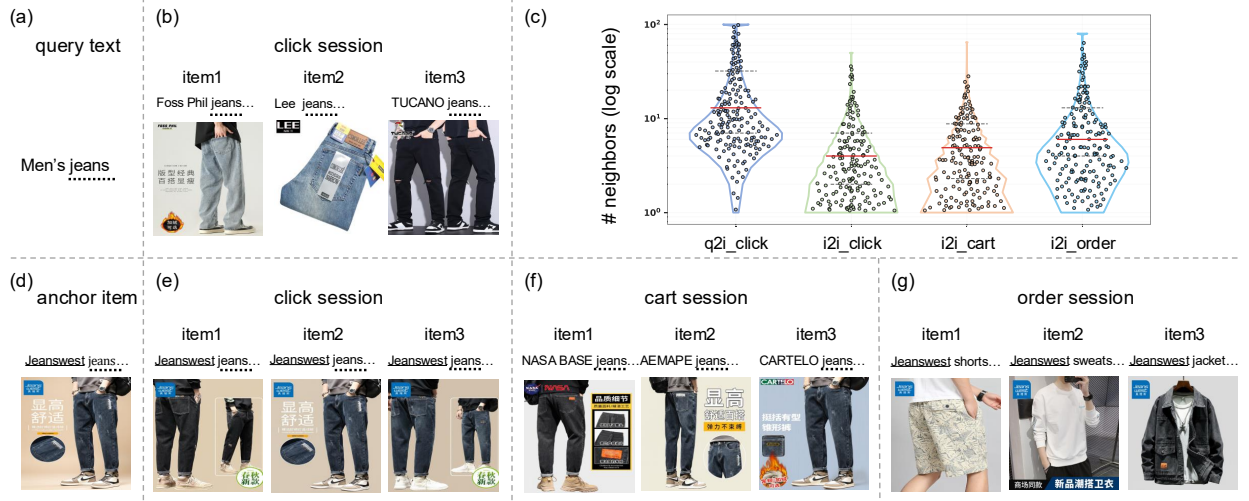


Figure 1: Illustration of heterogeneous collaborative signals. (a-b) Search-based q2i signals; (d-g) Behavior-based i2i signals within (e) click, (f) cart, and (g) order sessions; (c) Neighbor distribution across different collaborative signals.

To address these challenges, we propose the Multiplex Multi-modal Representation Model (MMRM), a unified framework aligning MLLMs with four diverse collaborative signals. By employing a shared backbone with task-specific tokens and projection layers, MMRM simultaneously learns from four signals without performance degradation, and efficiently generates four disentangled item representations in a single inference pass. Furthermore, we introduce a multiplex user representation strategy in ranking models, which derives task-specific user representations via search-based behavior sequence modeling using multiplex item representations. Integrating these user representations into corresponding task towers significantly enhances multitask ranking performance.

Our primary contributions are summarized as follows:

- We propose MMRM, a unified framework that effectively aligns MLLMs with diverse collaborative signals via a shared backbone and task-specific tokens and projection layers, enabling efficient generation of multiplex item representations.
- We propose a multiplex user representation strategy that derives task-specific user representations from multiplex item representations to boost multitask ranking performance.
- We demonstrate MMRM’s superiority through extensive offline and online experiments, and have deployed it online.

2 Method

2.1 Dataset Construction

2.1.1 Q2I Dataset. As shown in Fig. 2a, chronological user sessions typically progress from search queries to item clicks, cart additions, and orders, forming a complete behavior sequence. To align textual queries and multimodal items, we construct the q2i_click dataset using query-item clicks—the most pervasive and coarse-grained collaborative signal. To enhance contrastive learning, we randomly

sample a hard negative from the same parent category of the positive but a different sub-category. Our final q2i_click dataset comprises 0.3 billion diverse triplets collected over six months, capping each query at its top 150 interacted items (Table 1).

2.1.2 I2I Dataset. To complement broad q2i signals with fine-grained i2i relationships essential for multitask ranking, we construct three i2i datasets (click, cart, and order) via a graph-based approach (Fig. 2a). Using i2i_click as an example, we construct an item graph by connecting consecutive clicks within a 10-minute window, capturing strong collaborative relations (Table 1). After filtering noisy high-frequency items (promotions, takeaways, and coupons), nodes v_i and edges e_{ij} are weighted by their occurrences $f(v_i)$ and transition frequencies $g(e_{ij})$, respectively. To mitigate hot-item dominance, anchor items v_a are subsampled with probability:

$$P(v_a) = \sqrt{\frac{t}{f(v_a)}}, \quad (1)$$

where $t = 10^{-5}$ as per word2vec[16]. For each anchor, we sample up to 30 positive items v_p from its first- and second-order neighbors based on the transition probability:

$$P(v_p|v_a) = \frac{g(e_{ap}) + \sum_{j \in B_a} g(e_{aj})g(e_{jp})}{\sum_{j \in B_a} g(e_{aj}) + \sum_{j \in B_a} \sum_{k \in B_j} g(e_{aj})g(e_{jk})}, \quad (2)$$

where B_a denotes the neighbor set of v_a . Finally, we sample hard negatives v_n from the same sub-category as v_p , with probability[16]:

$$P(v_n|<v_a, v_p>, n \notin \{a, p\}) = \frac{f(v_n)^{3/4}}{\sum_{j \in U_{s(v_p)}} f(v_j)^{3/4}}, \quad (3)$$

where $s(v_p)$ is the sub-category of v_p , and $U_{s(v_p)}$ is the item set having the same sub-category as v_p .

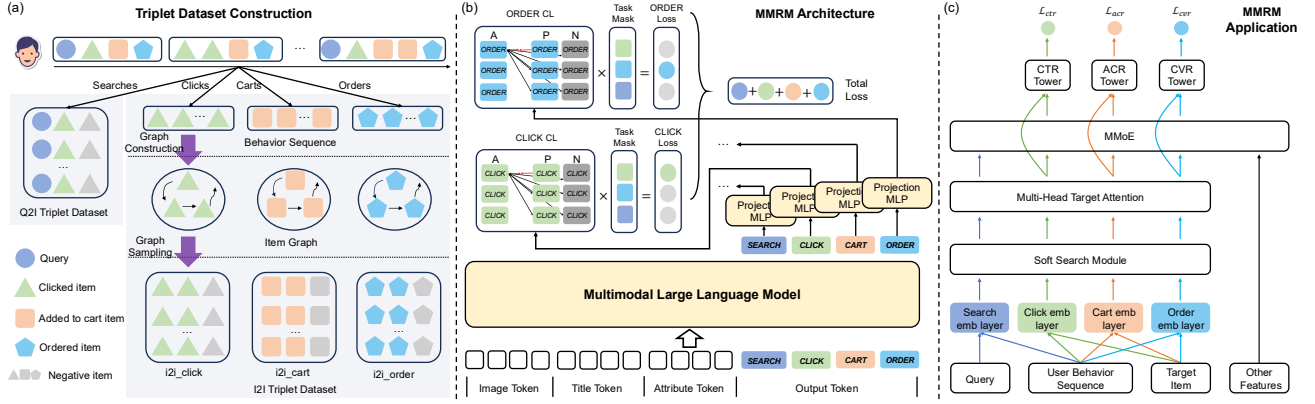


Figure 2: Overview of MMRM. (a) The pipeline for triplet dataset construction; (b) The MMRM architecture; (c) The MMRM application in multitask ranking models.

Table 1: Statistics of datasets for MMRM training and testing

Data type	Time period	Session window	Max neighbors	#Train Samples	#Test Samples
q2i_click	6 mo	-	150	0.3B	0.5M
i2i_click	6 mo	10 min	30	0.3B	0.5M
i2i_cart	1 yr	30 min	50	0.3B	0.5M
i2i_order	3 yr	7 d	100	0.3B	0.5M

The i2i_cart and i2i_order datasets follow the same generation process of i2i_click, differing only in configurations (Table 1). Unlike traditional CF-based methods, we capture fine-grained i2i relationships by constructing direct item graphs from sequential user behaviors. Additionally, hard negative sampling further enhances contrastive learning.

2.2 The Multiplex Multimodal Representation Model

2.2.1 Model Architecture. MMRM takes one q2i and three i2i datasets in a unified triplet format (Fig. 2b). Item inputs include an image, a title, and attributes (e.g., shop name, price), while queries are treated as special items without images and attributes. To disentangle tasks, four special tokens—[SEARCH], [CLICK], [CART], and [ORDER]—are appended to the inputs. Leveraging the autoregressive nature of MLLMs, the task-specific embedding for item i is obtained by projecting the last hidden state of special token t via a task-specific MLP:

$$\mathbf{r}_i^t = \text{MLP}^t(\mathbf{h}_i^t), t \in \{[\text{SEARCH}], [\text{CLICK}], [\text{CART}], [\text{ORDER}]\}. \quad (4)$$

2.2.2 Multitask Learning Objectives. During training, four datasets are mixed uniformly. For each task t , only its corresponding positive pairs are used for loss calculation, while items from other tasks act as in-batch negatives. The contrastive learning loss for task t is [22]:

$$\mathcal{L}_t^{\text{CL}} = -\frac{1}{\sum_{i=1}^B \mathbb{1}_t(i)} \sum_{i=1}^B \mathbb{1}_t(i) \log \frac{e^{\mathbf{r}_{ia}^t \cdot \mathbf{r}_{ip}^t / \tau^t}}{\sum_{j=1}^B e^{\mathbf{r}_{ia}^t \cdot \mathbf{r}_{jp}^t / \tau^t} + \sum_{j=1}^B e^{\mathbf{r}_{ia}^t \cdot \mathbf{r}_{jn}^t / \tau^t}}, \quad (5)$$

where B is the batch size; $\mathbb{1}_t(i)$ is an indicator function returning 1 if the i -th instance belongs to task t ; $\mathbf{r}_{ia}^t$, $\mathbf{r}_{ip}^t$, and $\mathbf{r}_{in}^t$ are anchor, positive, and negative embeddings, respectively, for the i -th instance from task t ; and τ^t is the temperature for task t . Finally, the total loss aggregates these task-specific losses with weights γ^t :

$$\mathcal{L}_{\text{total}}^{\text{CL}} = \sum_t \gamma^t \mathcal{L}_t^{\text{CL}}, t \in \{[\text{SEARCH}], [\text{CLICK}], [\text{CART}], [\text{ORDER}]\}. \quad (6)$$

2.3 The Multitask Ranking Model

As illustrated in Fig. 2c, given a query Q , a user behavior sequence S , a target item I , and other features O , we first extract a top- K relevant behavior subsequence S'_t using the multimodal embedding table specific to task t :

$$S'_t = \text{Soft-Search}(S, I/Q, K, t). \quad (7)$$

Then, we derive a task-specific user representation U_t using a Multi-Head Target Attention (MHTA) module (multi-head DIN[27]):

$$U_t = \text{MHTA}^t(S'_t, I/Q, t). \quad (8)$$

After that, all user representations and features O are processed by an MMoE module [14] to yield the task-specific hidden state X_t :

$$X_t = \text{MMoE}(U_{[\text{SEARCH}]}, U_{[\text{CLICK}]}, U_{[\text{CART}]}, U_{[\text{ORDER}]}, O, t). \quad (9)$$

At last, to predict for task t , X_t and the task-specific user representation U_t are jointly fed into an MLP-based tower network:

$$\hat{y}_t = \text{Tower}^t(X_t, U_t). \quad (10)$$

The overall multitask ranking loss is defined as a weighted sum of task-specific cross-entropy losses [15]:

$$\mathcal{L}_t^{rank} = \text{Cross-entropy}(y_t, \hat{y}_t), \quad (11)$$

$$\mathcal{L}_{total}^{rank} = \sum_t \lambda^t \mathcal{L}_t^{rank}, t \in \{ctr, acr, cvr\}, \quad (12)$$

where λ^t is the loss weight for task t .

3 Experiments

3.1 Experimental Setup

3.1.1 Datasets and Metrics. We evaluated MMRM’s multitask performance on four held-out test sets (0.5M samples each, Table 1). After models were trained, we derived embeddings used in test sets, performed ANN search tasks, and measured retrieval and ranking performance via F1@k[7] and NDCG@k[8], respectively. To evaluate ranking models, we collected 8-day JD search logs, and used the first 7 days for training (4B samples) and the last day for testing (2M samples). Following established practices, ranking performance was assessed via GAUC[27, 28].

3.1.2 Baseline Models. We compared three types of representation models: **Single** (task-specific models), **Vanilla-Multi** (multitask via task-specific prompts and a shared [EMB] token), and **MMRM** (multitask via task-specific tokens). Besides, we compared four types of SIM-based ranking models[19]: **SIM_{hard}(category)** (category-based behavior filtering), **SIM_{soft}(item_{e2e})** (filtering by end-to-end item embeddings; our online baseline[5]), **SIM_{soft}(item_{MMRM}[t])** (filtering by MMRM’s task t embeddings), and **SIM_{soft}(item_{MMRM}[ALL])** (multiplex filtering by all embeddings from MMRM).

3.1.3 Implementation Details. MMRM initialized from Qwen3-VL-4B-Instruct[1], training all parameters except the vision encoder. Training used Adam[9] (learning rate 2e-5, batch size 2048 via GradCache[4]), FlashAttention-2[3], $\tau^t = 0.05$, and $\gamma^t = 0.25$. Ranking models used AdamW[12] (learning rate 3e-5, batch size 4096) and $\lambda^t = 0.33$. All models were trained on 8 NVIDIA H800 GPUs.

3.2 Evaluations for Representation Models

3.2.1 Main Results. Evaluation results for representation models (Table 2) reveal three insights: (1) Single task models excelled only on their specific training tasks, highlighting task heterogeneity and the inability of a universal embedding to align all collaborative signals; (2) Vanilla-Multi underperformed all single-task baselines, indicating that task-specific prompts with a shared [EMB] token fail to disentangle tasks; (3) MMRM consistently outperformed all baselines, validating its architecture design. MMRM’s superiority over single task models likely stems from more diverse in-batch negatives during multitask training. Additionally, distinct task tokens enable MMRM to efficiently generate all four task embeddings in one forward pass, avoiding four separate passes required by baselines.

3.2.2 Ablation Study. Contrastive learning typically benefits from a large batch size[2], which is often bottlenecked by the high GPU memory consumption of modern MLLMs. To scale the batch size, we employed GradCache[4], which splits the backpropagation of the contrastive loss into two stages and breaks the model update into memory-efficient sub-updates. This enabled us to train the

4B-parameter MMRM with a batch size of 4096 on a single H800 machine. Our ablation study on the batch size (Table 3) confirms that larger batch sizes consistently boost MMRM’s performance.

3.3 Evaluations for Ranking Models

Table 4 presents evaluation results for ranking models. Our online baseline, **SIM_{soft}(item_{e2e})**, outperformed **SIM_{hard}(category)**, validating the soft search strategy. However, **SIM_{soft}(item_{e2e})** relies on a single end-to-end embedding table for behavior filtering and user modeling. This risks embedding entanglement, thereby limiting performance in multitask scenarios. By incorporating an additional multimodal embedding table for soft search, **SIM_{soft}(item_{MMRM}[t])** achieved significantly better results across all three tasks. Ultimately, utilizing all four multimodal embedding tables for task-specific user modeling, our proposed **SIM_{soft}(item_{MMRM}[ALL])** attained the best performance, demonstrating the effectiveness of multiplex multimodal representations in multitask ranking.

3.4 Online A/B Test

In a week-long online A/B test on the JD e-commerce search engine, our proposed **SIM_{soft}(item_{MMRM}[ALL])** outperformed the online baseline **SIM_{soft}(item_{e2e})**, improving user click-through (UCTR), add-to-cart (UACR), and conversion (UCVR) rates by 0.42%, 0.37%, and 0.35%, respectively. Given the massive traffic of the platform, even a 0.10% lift in these metrics yields substantial revenue growth, making these gains highly significant. Consequently, both MMRM and the proposed ranking model have been fully deployed online.

4 Conclusions

In this paper, we proposed the Multiplex Multimodal Representation Model (MMRM) to integrate multimodal information into ranking models. By coupling a shared backbone with task-specific tokens and projection layers, MMRM simultaneously aligns with four graph-derived collaborative signals without performance degradation, generating comprehensive multiplex item representations in a single inference pass. Furthermore, we introduced a multiplex user representation strategy leveraging multiplex item representations, boosting downstream multitask ranking performance. Extensive offline and online experiments validated the effectiveness of MMRM and the proposed ranking model.

5 Presenter Biography

Zhen-Lin Chen is an applied scientist at JD.COM, focusing on research and development of large-scale search and recommendation systems to improve user online shopping experiences. He received his doctoral degree from the Institute of Computing Technology, Chinese Academy of Sciences. His research mainly focuses on natural language processing and information retrieval.

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Table 2: Evaluation results for representation models. Bold and underline denote the best and second-best results, respectively.

Task		F1@5				NDCG@5			
		q2i_click	i2i_click	i2i_cart	i2i_order	q2i_click	i2i_click	i2i_cart	i2i_order
Single	q2i_click	0.1985	0.2943	0.2679	0.0675	0.6356	0.7289	0.6655	0.4752
	i2i_click	0.0818	<u>0.3254</u>	0.2981	0.0831	0.6129	<u>0.7367</u>	0.6752	0.4976
	i2i_cart	0.0238	0.3237	<u>0.3013</u>	<u>0.0908</u>	0.5948	0.7351	<u>0.6763</u>	0.5070
	i2i_order	0.0087	0.2989	0.2809	0.0904	0.5579	0.7293	0.6710	<u>0.5224</u>
Vanilla-Multi		0.1593	0.2987	0.2748	0.0676	0.6127	0.7295	0.6702	0.4875
MMRM		0.2055	0.3276	0.3037	0.0934	0.6392	0.7371	0.6779	0.5249

Table 3: Ablation study on batch size. GradCache was used for batch sizes > 1024, and images were omitted for acceleration.

Task		F1@5			
		q2i_click	i2i_click	i2i_cart	i2i_order
Batch size	256	0.1874	0.3216	0.2991	0.0882
	512	0.1937	0.3223	0.2984	0.0895
	1024	0.1980	0.3239	0.2991	0.0907
	2048	0.2007	0.3255	0.3004	0.0916
	4096	0.1997	0.3251	0.3021	0.0919

Table 4: Evaluation results for ranking models

Task		GAUC		
		CTR	ACR	CVR
SIM _{hard} (category)		0.6816	0.7051	0.6733
SIM _{soft} (item _{e2e})		0.6866	0.7101	0.6783
SIM _{soft} (item _{MMRM} [SEARCH])		0.6947	0.7160	0.6807
SIM _{soft} (item _{MMRM} [CLICK])		0.6928	0.7145	0.6839
SIM _{soft} (item _{MMRM} [CART])		0.6941	0.7124	0.6830
SIM _{soft} (item _{MMRM} [ORDER])		0.6934	0.7127	0.6814
SIM _{soft} (item _{MMRM} [ALL])		0.7037	0.7190	0.6875

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